

AI Tools Used

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	2 Marks

AI Tools Used:

The table below documents the specific AI tools and platforms used throughout the development of CaptionAI, detailing how each tool was applied and the tangible benefit it provided to the workflow and final product.

S.No	AI Tool / Platform Name	Specific Use Case / Application	Project Phase Applied	Impact / Benefit
1	ChatGPT (OpenAI)	Brainstormed and finalized the CaptionAI project idea; prepared the project proposal and planning.	Ideation & Planning Phase	Helped quickly select and scope a viable project idea.
2	ChatGPT (OpenAI)	Assisted in writing the Python Flask backend, including routes, Groq API integration, and SQLite database implementation.	Development Phase	Reduced backend development time and improved code quality.
3	ChatGPT (OpenAI)	Generated responsive HTML, CSS, and JavaScript code for the login, registration, dashboard, history, and caption generation pages.	Design / Frontend Development Phase	Simplified and sped up frontend UI creation.
4	ChatGPT (OpenAI)	Identified and fixed Flask routing errors, SQLite issues, template rendering problems, and API integration errors.	Testing & Debugging Phase	Simplified debugging and enabled faster issue resolution.
5	ChatGPT (OpenAI)	Prepared project documentation including the DFD, Solution Requirements, Technology Stack, Problem–Solution Fit, Empathy Map, and Customer Journey Map.	Documentation Phase	Helped prepare structured, well-organized project documentation.
6	Groq API (LLaMA 3.3-70B Versatile)	Used as the core AI engine to generate platform-specific captions, relevant hashtags, and engaging call-to-action suggestions in response to user prompts.	Core Application Feature (Runtime)	Delivered fast, high-quality AI-generated content, improving the overall user experience.

Summary

The project successfully utilized AI tools for both development and application functionality. ChatGPT supported planning, coding, debugging, and documentation, while the Groq API powered the core feature of AI-based caption generation. These tools helped improve productivity, reduce manual effort, and enhance the overall

quality of the project.